INORGANIC CUMULATIVE EXAM

December 5, 1992

Group theory can be used to predict the number of vibrational bands that will appear in the IR or Raman spectrum of a compound if the point group for the molecule is known. Conversely, the observed number of bands in the spectrum can frequently be used to decide which of two possible structures is most likely correct for a molecule.

1) What is the general condition required for a particular fundamental vibrational mode to be observed in the transmission IR and in the Raman spectrum?

2) The molecule $Re(CO)_4Cl_2$ can potentially show two geometric isomers. What is the total number of vibrational modes possible for this molecule?

3) The CO stretching frequencies for $Re(CO)_4Cl_2$ would be well separated from the other modes for the molecule, determine the number of IR and Raman active bands and the polarization characteristics for the Raman bands for the two isomers of $Re(CO)_4Cl_2$.

4) Frequently, the experimental vibrational spectrum of a molecule will appear to have more observed bands than predicted by the group theory methods used to answer questions above. Give as many reasons as you can to account for this observation.